SHIV NADAR UNIVERSITY

CSD 207 Object Oriented Programming in Java

Instructors

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Semester of Implementation: Monsoon 2018

PART A: COURSE IDENTIFIERS

Fill both columns if the course is a joint offering and/or is cross-listed with different codes.

1. School	**School of Engineering**
2. Department	**Computer Science and Engineering**
3. Course Code	**CSD207**
4. Course Title	**Object Oriented Programming in Java**
5. Credits (L:T:P:C)	**3:0:1:4**
6. Contact Hours (L:T:P)*	**3:0:2**
7. Prerequisites	**CSD101**
8. Major Core for**	**CSE**
9. Major Elective for	**None**

***Per student per week. For example, a 1.5 credit CCC that runs for a half-semester with only lectures would have contact hours 3:0:0. A course with 3 lecture hours and 1 tutorial hour and a single 3-hour lab per week would have contact hours 3:1:3.
**Give Major and Year.

PART B: OBJECTIVES AND PRACTICE

10. Course Summary

This course includes the introductory and advanced concepts and implementation of the Object Oriented Paradigm using the Java programming language. Topic would include Introduction, Elementary Programming, Selections, Loops, Methods, Arrays, Strings, Objects and Classes, Inheritance and Polymorphism, GUI Basics and Components, Graphics, Exceptions, Abstract Classes and Interfaces, Event-Driven Programming, Binary I/O, Recursion, java database programming and Java Collections Framework.

11. Course Aims

This course will enable the students to understand the concepts of object oriented programming using java, and apply its structures and techniques in various programming problems.

12. Learning Outcomes

On successful completion of the course, students will be able to achieve the following:

1. Ability to understand, write and execute java programs using various input and output mechanisms.
2. Understand and use constants, variables and data types used in java.
3. Understand and apply selection and loop statements.
4. Understand and apply the concepts of methods.
5. Understand and apply string-handling mechanisms.
6. Understand object oriented programming concepts.
7. Understand and use abstract classes and interfaces.
8. Understand use Inheritance and Polymorphism.
9. Understand and create Graphical user interface.
10. Understand and perform event driven programming.
11. Understand and apply exception handling.
12. Understand and apply the components of Java Collections Framework.
13. Understand and apply java database programming.

13. Curriculum Content

Week 1

Introduction to Computers, Programming, and Java

Elementary Programming

Week 2
Selections

Week 3
Loops

Week 4
Methods

Week 5
Single-Dimensional Arrays
Multidimensional Arrays

Week 6
String handling

Week 7
Object oriented programming: Classes, Objects, Inheritance and Polymorphism

Week 8

Object Oriented Programming – continued.

Week 9-10
GUI

Week 11
Event-Driven Programming

Week 12
Exception Handling

Week 13-14
Java collection Framework: Lists, Stacks, Queues, and Priority Queues

Week 15
Java Database programming

Week 16-17
Course Review

14. Teaching and Learning Strategy

All materials (ppts, assignments, labs, etc.) will be uploaded in BlackBoard. Refer to the modules in Blackboard and weekly schedule for details. Students will use NetBeans IDE with latest version of JDK to write the programs.

PART C: ASSSESSMENT.

15. Evaluation Instruments

Evaluation Instrument	Weightage
Mid-term Exam	20%
Assignments/Quizzes	20%
Laboratory	20%
Project	10%
End-term Exam	30%

Mapping of Learning Outcomes to Assessment Strategy

For details mapping, please refer to each course contents in Blackboard for details.

16. Bibliography

Textbook: Introduction to Java Programming, by Y. Daniel Liang, Comprehensive version, 7th edition.

Reference book: Java: The Complete Reference, by Herbert Schildt, 7th edition.

Grading Policy

Scores obtained in all the components of Evaluation shall be totaled and the final score will be converted into letter grade. Students taking this course to improve their earlier grade are reminded that this grade replaces the previous grade (regardless of whichever is higher).

Grading Scheme

Grade	Range
A	90 and above
A-	80 to 89.99
B	70 to 79.99
B-	60 to 69.99
C	50 to 59.99
C-	45 to 49.99
D	40 to 44.99
F	39.99 and below

Make up Policy

No make-up will be granted for any component of evaluation unless:

- You have emailed the course instructors with necessary reason(s) as soon as possible before the date of evaluation and a make-up date has been agreed upon.
- Unforeseeable circumstances prevent you from keeping the original date.

In either case, you may need to submit supporting documents. The decision of the Course Instructor in all these matters shall be final.

Cheating Policy

Any attempt to pass of someone else work as your own or to assist someone else in such an effort is considered cheating. Please familiarize yourself with SNU policy on Academic Malpractice and Plagiarism.

Attendance and Participation

The nature of the class makes participation essential.

- Make sure you sign your name in each class.
- Missing Classes will be subjected to penalty as per SNU rules unless there is an emergency. If you become seriously ill or have a family emergency, it is your responsibility to communicate with the Professors as soon as you are able, preferably before class. The Professors reserve the right to require documentation.
- Please familiarize yourself with the University policy on Attendance.

Course Policies

- Maintain a notebook and keep notes for each and every lecture.
- No social Networking/non relevant sites should be opened during the class.
- No food and drinks allowed in the class.
- Course Materials will be regularly posted.
- Bonus points may be given during the semester in the form of quizzes and assignments.
- See Announcement Section regularly for other information.

Electronic Device Policy

All Electronic items including mobile phones/tablets will not be allowed with the students during lectures and labs. Students bringing them due to unavoidable circumstances will have to keep it switched off in their bag and the bag should be kept at a designated place, and certainly not with the students. Allowing laptops will be at the discretion of the instructor, based on the needs of the course.

If a student is found violating this rule, following action will be taken against the student:

On First Instance:

- No attendance will be given to the student for that lecture/lab.
- Written apology will be taken from the student.

On Second and future Instances:

- No attendance will be given to the student for that lecture/lab.
- Penalty of two points (on a 100 point scale) per incident for the student. Student will have to immediately keep away the electronic item following the incident.